

ORF309/MAT380

Introduction & Course Information

Ioannis Akrotirianakis

Agenda for today

- A general overview of the course
- Logistics – Course Policies
- Intuition and Probability
 - Some applications of Probability (tell me your experiences with probability)
 - Determinism and Randomness
 - Two perfectly intuitive examples

Course components

- **Lectures:**
 - **Mon/Wed/Fri: 9:35 – 10:25 AM**
 - *Lecture notes on Canvas (by Professor Ramon van Handlel)*
 - *Lecture slides, Precepts, PSETs on Canvas*
- **Precepts** (start next week):
 - **Mondays:** [P01](#) 3:30-4:20pm ; [P02](#) 7:30-8:20pm
 - **Tuesdays:** [P03](#) 3:30-4:20pm; [P04](#) 7:30-8:20pm
 - [Review of the lectures and material needed for PSETs](#); **please attend!**

Course components

- **PSETs**
 - **Nine (9)** PSETs due on **Wednesdays before 11:59PM** (1minute before midnight)
 - They will be posted under **Assignments on Canvas** and solutions are submitted to **Gradescope**.
 - **Collaboration is encouraged!** But each student must submit their own solution on Gradescope.
 - **Extension Policy:** Requests for extensions will **not** be accepted without valid documentation. For any **health-related extensions**, you must include your **Residential College Dean** in your request. After the release of the solutions, no HW will be considered.
 - We will **drop automatically the lowest grade** from the final grade computation
 - **Solutions** will be posted soon after the deadlines (study them carefully).

Course components

- There will be **2 midterm exams**:
 - Midterm 1: Week 6 Midterm 2: ~Week 10
- **Final exam**: after Dean date
- **Grading**:
 - PSETs: 20%; Midterm1: 20%; Midterm2: 20%; Final: 40%
- **Details can be found on Canvas; please read carefully!**

Generative AI (GenAI) policy

- 1) The use of GenAI is NOT allowed for directly solving PSETs or for the 2 Midterms and the Final exam.
- 2) You may use GenAI to **analyze past assignments** (i.e., assignments you have already submitted for grading, or past exams we will provide you for practice). For example, you could use it to explore different solutions to PSETs that have been submitted or past exams.
- 3) You may also use any GenAI to explain concepts; specifically, you can use **\explain <topic>** as a prompt but without any further prompting.
- 4) In case you do use GAI, **you must submit the prompt and output from the language model as part of your assignment submission**. Any other use of GenAI beyond what is described here will not be allowed for this course.
- 5) As always, remember that you are bound by the **Princeton Honor Code**, and violations can have serious consequences.

Office hours

- **Ioannis:** TBD
- **Yucheng** → Monday 5:30pm - 7:30pm, Sherrerd 123
- **Qinxin** → Tuesday 1:30pm - 3:30pm, Sherrerd 122
- **Ximing** → Wednesday 2:00pm - 4:00pm, Sherrerd 123
- **Francesco** → Wednesday 4:00pm - 6:00pm, Sherrerd 123
- **Kyle** → Friday 2:00pm - 4:00pm, Sherrerd 107
- **UCAs:** TBD

Also post any questions on EdDiscussion!

What is Probability and (some of) its applications

- **Probability theory** = “the study of Randomness” → Randomness exists everywhere
- **Economics/Finance:**
 - Stock or Bond prediction heavily relies on probability theory (what is the probability that **NVIDIA** will be **up** or **down** tomorrow)
 - Probability of **rare events** happening form the basis of **insurance companies**
- **Physics:**
 - **Temperature/Pressure** are consequences of the random motion of atoms
 - **Quantum Mechanics** which describes the behavior of particles not with certainty but with probabilities based on their “*wave functions*”

Roughly speaking, the wave function of a particle helps describes the probability to find a particle at certain location at a certain time.

What is Probability and (some of) its applications

- **Computer Science:**
 - **Randomized algorithms** are used to solve computationally expensive problems (e.g., genetic algorithms)
- **Biology and Medicine**
 - Probability can help us understand and model the **spread of diseases** (e.g., flu, COVID)
- **Chemistry**
 - **Chemical reactions** happen when molecules randomly meet ("human reactions" are governed by the similar randomness too...)
- **Mathematics**
 - It is believed that **Prime numbers** behave like random numbers

The usefulness of Probability Theory:

Probability theory does not stop the randomness in the world, but rather gives us a powerful way of **thinking and reasoning** to see the **patterns** within it, and allows us to ***predict, survive and thrive***

What is randomness

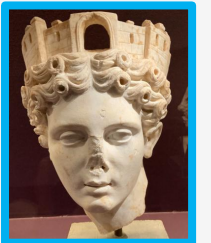
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- Life can be thought as a series of experiments (or phenomena)
- **Deterministic** experiment
 - We can predict the outcome of the system every time under the same conditions exist (same conditions → same (known) outcome) – **Example: Drop a ball from the same place.**
- **Random** experiment
 - The outcome of the experiment varies and is randomly selected (e.g., by **Tyche**) even when the conditions have not changed. **Example: Flip a coin or roll a die**
- **Humanity's objective:** develop a mathematical reasoning framework about randomness



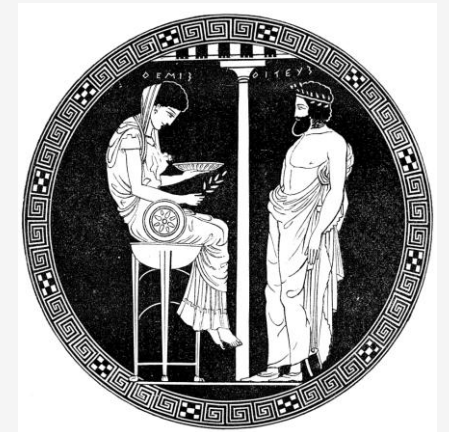
Tyche

In ancient Greece, **Tyche** was responsible for the management of uncertainty and randomness.

She kept all her equations secret!



- Daughter of Zeus and Aphrodite
- She governed the fortune, prosperity and fate of cities.
- She operated in a capricious and random manner.
- Perhaps she the reason why the ancient Greeks did not study Probability although they were aware of randomness and uncertainty, but they let it to the gods via the **oracle of Delphi**.



Probability seems intuitive – but be very careful with your intuition!

- We will rely on **our intuition** throughout the course to build our theory
 - **But we should be careful... not all intuitions lead to strong mathematical foundations**
 - Common intuition says that “**Probability that an event E happens**” should be “**the *fraction* of outcomes in which E occurs**”
- Mathematically: $Prob(E) = \frac{\text{\# of outcomes where } E \text{ occurs}}{\text{\# of all possible outcomes of event } E} \quad (= Freq(E))$ **Laplacian definition of probability (1812)**
- Is this useful intuition?? Can it lead us to a strong foundation of Probability??

Example with not unique solution

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 - $\Omega = \{2H, 1H \& 1T, 2T\}$ → John says: all outcomes are **equally probable**
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- **Alice’s solution:** She finds a slightly different set of **possible outcomes** (she cares about the **order** of $H\&T$)
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- **John’s solution:** Finds all the set of all possible outcomes of the experiment
 - $\Omega = \{2H, 1H \& 1T, 2T\} \rightarrow$ **John is wrong:** all outcomes **are NOT** equally probable
“1H & 1T” should twice more probable than the other two
 - $P(1H \& 1T) \neq 1/3$
- **Alice’s solution:** She finds a slightly different set of possible outcomes (the order of $H\&T$ is important)
 - $\Omega_2 = \{HH, HT, TH, TT\} \rightarrow$ **Alice is correct:** all outcomes are **equally probable**
 - $P(1H \& 1T) = P(\{HT, TH\}) = 1/2$

*In 1750s this was the right answer
supported by d’Alabert in his encyclopedia*

This is right

Some thoughts

- The Intuitive Definition of Probability we gave:

$$Prob(E) = \frac{\# \text{ of outcomes where } E \text{ occurs}}{\# \text{ of all possible outcomes of event } E} \quad (= Freq(E))$$

- **cannot be correct** because in that case ***both Solutions would be correct***
- It relies too much on intuition! → It cannot model all forms of reality...
- We need a **mathematical way to study random phenomena** → This gives rise to **Probability Theory** → We will be able to model random phenomena and compute their probabilities accurately

A brief history of Probability

- **Gambling** has been a driving force in the development of a rigorous mathematical theory of Probability



- Notable **pioneers** of Probability:

- 1654 {
 - **Chevalier de Méré** (Nobleman and **gambler**)
The game: at least one double 6 in 24 throws of 2 dice
 - **Blaise Pascal** (Mathematician, **gambler** and very religious person)
 - **Pierre de Fermat** (**Lawyer** and (amateur) mathematician)
- 1684 {
 - **Jacob Bernoulli** (Mathematician)
(derived the probability of obtaining k successes in n trials $\rightarrow$ Binomial distribution)
- 1933 {
 - **Andrey Kolmogorov** (Mathematician; founder of Probability Theory)
(proposed the 3 fundamental axioms)

Objectives of the course

- Introduce the **basic mathematical principles of Probability**
- How to use our **intuition as guide**, but always **check if it correct**
- Extend our knowledge of Probability to **Random Processes**
 - Random processes deal with the evolution of randomness in *future time*
 - **Examples:** Poisson processes, Random walks, Brownian motion, Markov chain
- The course is designed for a **broad audience** and is not following a theorem-proof style. Nevertheless, **it requires rigorous mathematical thinking!**
- At the end: You will be able to reason about randomness and think probabilistically
